



GNC Engineer

Astrolab is The Lunar Operations Company. We design and build the systems and hardware the lunar economy will come to depend on — rovers, power, and data networks for sustained operations on the Moon. We are focused on the problems that must be solved before the next chapter of human exploration can **scale**. The Moon does not forgive untested assumptions. The people who work here know that, and they wouldn't have it any other way.

As a GNC engineer at Astrolab, you will be responsible for designing, analyzing, and testing Guidance, Navigation, and Control algorithms for our lunar rover fleet. You will support development and maturation of GNC for crewed, semi-autonomous, and fully autonomous rover operations. GNC engineers own hardware selection and characterization, algorithm development and tuning, algorithm and integrated performance analysis, and verification and validation for the GNC subsystem. You will work with simulated environment as well as test vehicles to design, mature, and prove GNC functionality and performance.

Note: Some travel may be required for this position

Department

Software Engineering

Employment Type

Full Time

Location

Hawthorne, California

Workplace type

Fully remote

Compensation

\$130,000 - \$225,000 / year

This role's hiring manager:



Denise Brown

[View Denise's Profile](#)

What you'll do:

- Define GNC subsystem and algorithm requirements
- Design, analyze, develop, tune, verify, and validate rover GNC algorithms and support conversion of algorithm prototypes into FSW
- Contribute to the development of the GNC simulation
- Work with avionics and other engineering teams to select GNC sensors such as IMUs, star trackers, cameras, LiDARs, etc.
- Support GNC sensor characterization, calibration, and testing
- Support development of integrated hardware/software GNC subsystem testbeds
- Develop GNC algorithm documentation and work with vehicle software team to convert prototypes into flight code, including supporting unit and integrated testing of flight implementations

- Work with vehicle software team, avionics team, mobility team, and integration and test team to verify GNC subsystem implementation and performance
- Support field testing of GNC software using test vehicles in a variety of lighting and terrain conditions
- Develop GNC analysis and test plans and perform GNC analysis and testing on hardware in the loop testbeds
- Support vehicle integration and testing of GNC functions
- Develop vehicle level calibration plans and support vehicle level calibrations

What you'll bring:

Basic Qualifications

- Bachelor's degree in a Science, Technology, Engineering, or Mathematics (STEM) related field, or an equivalent combination of education and experience
- 3+ years of professional experience developing Guidance, Navigation, and Control (GNC) systems
- Experience with Python and familiarity with C++

Preferred Qualifications

- Experience designing, tuning, and testing Extended Kalman Filters for vehicle state estimation.
- Experience designing, tuning, and testing vehicle control algorithms (e.g., high-gain antenna pointing)
- Experience building, testing, and validating GNC simulation tools
- Experience performing GNC performance analysis, including Monte Carlo analysis
- Experience supporting GNC development from initial concept through flight operations
- Experience collaborating across multidisciplinary engineering teams to mature GNC designs into flight software
- Strong technical writing and verbal communication skills

Benefits and Perks

- Join a team of best-in-class engineers building the foundation of planetary surface exploration
- Equity ownership in the company
- Comprehensive health benefits, including medical, dental, vision, and mental health support
- 401(k) plan with company match
- Flexible PTO and parental leave
- Home office set up reimbursement
- Weekly lunch stipend, plus complimentary snacks and beverages on-site
- Once a month social hour on-site with food and drinks

About Astrolab

Having a strong, inclusive, and inspired team is essential to accomplishing our bold mission. Employees are our biggest asset, and we are deeply committed to them and our values. At Astrolab we...

Demonstrate and inspire trust. Interactions are direct, honest, and respectful. We are fair and transparent with what we do and why.

Thrive in balance. Balance gives us perspective and keeps us motivated and inspired. We commit to rewarding work and a fulfilling personal life.

Seek diverse perspectives. We ask questions, engage in productive debate and offer alternatives to ensure our solutions and products are robust and a reflection of our collective best.

Move with purpose. Quality and reliability are accomplished through efficient ideation and iteration. We act with a sense of urgency and deliver what we promise, always with the big picture in mind.

Make an impact. We intentionally move our community and industry into the future through our work, practices, and actions.

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Our Hiring Process

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Astrolab is proud to be an equal opportunity workplace. We are committed to equal opportunity regardless of race, color, ancestry, religion, gender, gender identity, parental or pregnancy status, national origin, sexual orientation, age, marital status, disability, or veteran status. If you need assistance and/or reasonable accommodation due to a disability during the application or the recruiting process, please send a request to careers@astrolab.space. We strive to remove unconscious and unnecessary barriers to build the best team. While we've outlined what an ideal candidate could look like, we know that you may bring something unanticipated and essential to the team. If you're reading this and can see yourself contributing, please register below.

[Register Your Interest](#)

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